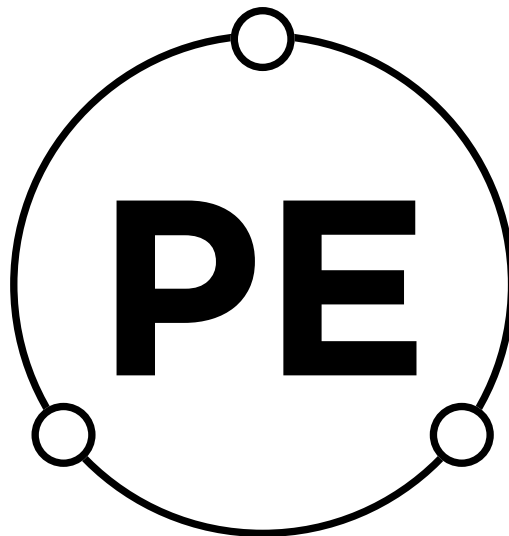


POUNAMU ELECTRONICS  
RE\_SWC  
USER GUIDE

*Rev 3.2*

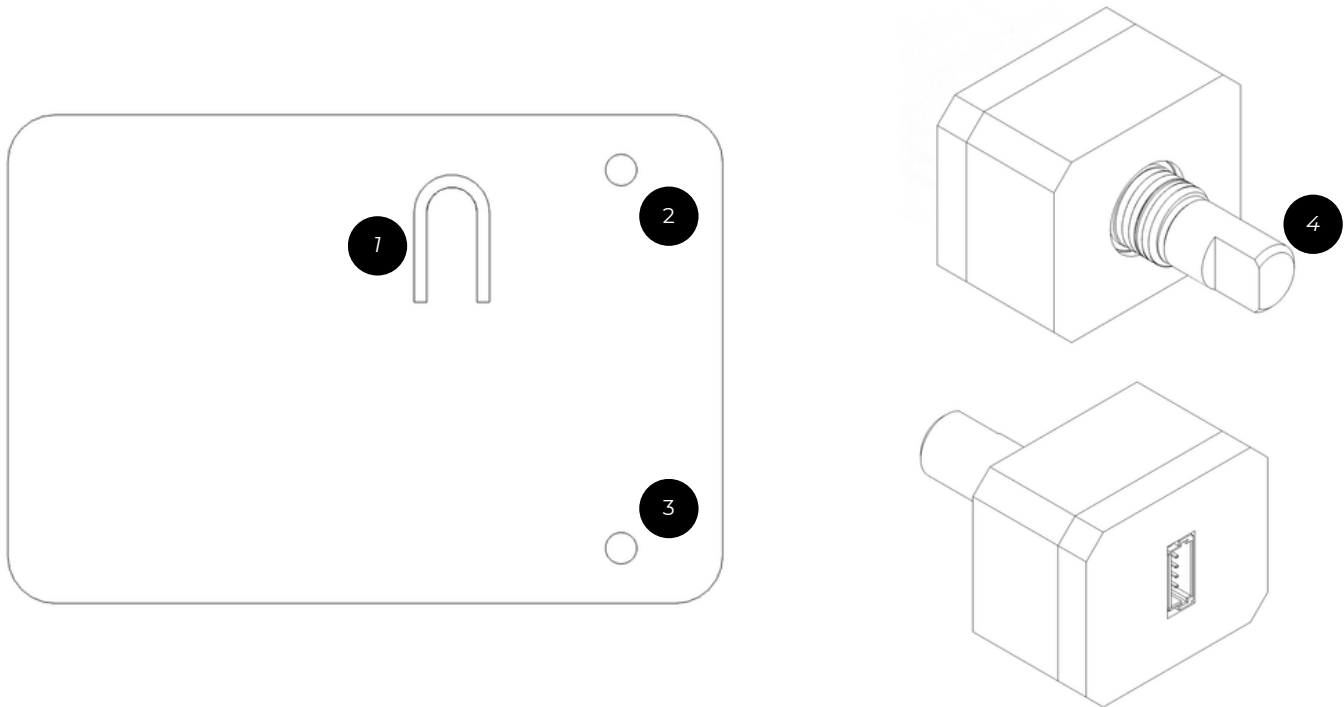


**POUNAMU  
ELECTRONICS**

# Contents

|    |                                           |
|----|-------------------------------------------|
| 3  | <u>OVERVIEW</u>                           |
| 4  | <u>ASSEMBLY</u>                           |
| 5  | <u>CONFIGURATION - HEADUNIT SELECTION</u> |
| 6  | <u>HEADUNIT LEGEND</u>                    |
| 7  | <u>FUNCTIONS</u>                          |
| 8  | <u>VOLUME KNOB DIMENSIONS</u>             |
| 9  | <u>TROUBLESHOOTING</u>                    |
| 10 | <u>RE_SWC SPECS</u>                       |
| 11 | <u>DISCLAIMERS</u>                        |
| 12 | <u>COMPLIANCE</u>                         |
| 13 | <u>DOCUMENT REVISION HISTORY</u>          |

# OVERVIEW



**1. UPDATE BUTTON**

**2. WHITE STATUS  
LIGHT**

**3. RED POWER  
LIGHT**

**4. DIA 6mm  
ENCODER SHAFT**

# ASSEMBLY

## USB HID

- Connect the RE\_SWC kit to your device with a USB C cable (must support data & **5V power**)
- The RED power indicator on the RE\_SWC will illuminate and stay on

## Others

- Solder the RED wire from the 2-pin connector on the RE\_SWC kit to your headunit's steering wheel control input pin
- Solder the BLACK wire from the 2-pin connector into your harness if your headunit requires a reference for the SWC input (sometimes labelled ST REMOTE GND)
- Power the kit with a USB C cable.
  - NOTE: This kit requires **5V USB power only**
- The RED power indicator on the RE\_SWC will illuminate and stay on

## Mounting the knob

- Drill a pilot hole between 1-3mm in diameter
- Widen the hole to 7mm
- Mount the knob into the hole and secure with the included washer and nut

## CONFIGURATION

### HEADUNIT SELECTION

- *Ensure RE\_SWC is not powered*
- *Press and hold the volume knob. Connect power to the RE\_SWC while keeping the volume knob held*
- *Release the volume knob after the WHITE status light is illuminated*
- *Press the volume knob once. The WHITE status light will flash once. Headunit number 1 is now programmed. Repeat this step until the desired headunit number is reached - refer to the headunit legend*
- *When the desired headunit number is entered, press and hold the volume knob. The WHITE status light will flash. Release the volume knob. The RE\_SWC will flash x times where x is the headunit number entered*

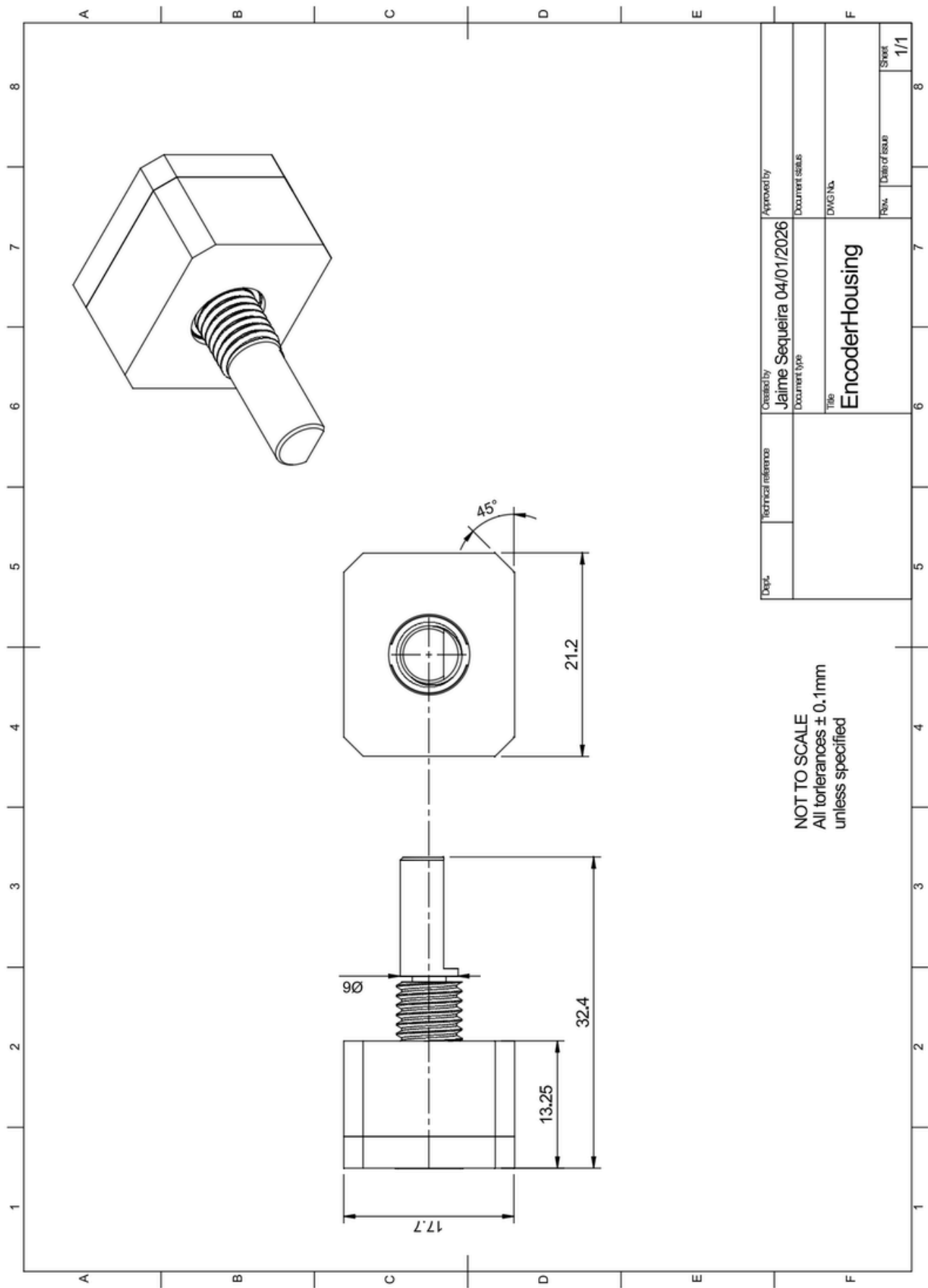
# HEADUNIT LEGEND

| <b>HEADUNIT NUMBER</b> | <b>HEADUNIT TYPE / BRAND</b> |
|------------------------|------------------------------|
| <b>1</b>               | <i>GENERIC RESISTIVE</i>     |
| <b>2</b>               | <i>JVC</i>                   |
| <b>3</b>               | <i>KENWOOD</i>               |
| <b>4</b>               | <i>ALPINE</i>                |
| <b>5</b>               | <i>PIONEER</i>               |
| <b>6</b>               | <i>USB HID</i>               |
| <b>7</b>               | <i>SONY</i>                  |

# FUNCTIONS

| <b>INPUT</b>                   | <b>GENERIC<br/>RESISTIVE</b>       | <b>JVC</b>                | <b>KENWOOD</b>            | <b>ALPINE</b>             | <b>PIONEER</b>            | <b>USB HID</b>            | <b>SONY</b>               |
|--------------------------------|------------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| <b>Volume Knob<br/>CW</b>      | <i>Any</i>                         | <i>Volume +</i>           | <i>Volume +</i>           | <i>Volume +</i>           | <i>Volume +</i>           | <i>Volume +</i>           | <i>Volume +</i>           |
| <b>Volume Knob<br/>CCW</b>     | <i>Any</i>                         | <i>Volume -</i>           | <i>Volume -</i>           | <i>Volume -</i>           | <i>Volume -</i>           | <i>Volume -</i>           | <i>Volume -</i>           |
| <b>Button Short<br/>Press</b>  | <i>Any</i>                         | <i>Mute</i>               | <i>Mute</i>               | <i>Mute</i>               | <i>Mute</i>               | <i>Mute</i>               | <i>Mute</i>               |
| <b>Button Long<br/>Press</b>   | <i>Any</i>                         | <i>Next Track</i>         | <i>Next Track</i>         | <i>Next Track</i>         | <i>Next Track</i>         | <i>Next Track</i>         | <i>Next Track</i>         |
| <b>Button Double<br/>Press</b> | <i>Enter<br/>Learning<br/>Mode</i> | <i>Previous<br/>Track</i> | <i>Previous<br/>Track</i> | <i>Previous<br/>Track</i> | <i>Previous<br/>Track</i> | <i>Previous<br/>Track</i> | <i>Previous<br/>Track</i> |

# VOLUME KNOB DIMENSIONS





# TROUBLESHOOTING

- **The RED power light is off**
  - *Ensure you are powering the kit with 5V USB power*
- **The RED power light is on but the WHITE status light does not flash when I use the knob**
  - *Ensure the 5-pin connector harness for the knob is seated correctly and inserted fully into the knob and the kit*
- **The WHITE status light flashes when I turn the knob but my headunit does nothing**
  - *Ensure you have set the kit in the correct headunit mode*
  - *Ensure the RED wire from the 2-pin connector is soldered properly into your headunit's harness*
- **My headunit shows the volume increasing but the volume of my music doesn't change in USB HID mode**
  - *Some Android headunits do not route USB volume commands to the correct hardware. Revert to using the Resistive SWC mode with Key1 / Key 2 input wire instead*

# RE\_SWC SPECS

## **Input**

Power: 5V / 100mA USB-C

SWC Header: **5V max.** Do not connect to 12V signal

## **CONNECTORS**

VOLUME KNOB: JST SHR-05V-S-B

SWC OUT: JST SHR-02V-S-B

## **MCU:**

WCH CH32X035F8U6

## **Digital Rheostat:**

MCP4131-104E/SN

SPI @ 250kHz

# DISCLAIMERS

- *Pounamu Electronics is not affiliated or associated with any headunit manufacturers or brands*
- *Pounamu Electronics is not affiliated or associated with the USB Implementers Forum (USB-IF)*

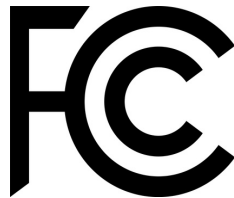
# COMPLIANCE

## ***FCC Compliance Statement***

*This device complies with Part 15 of the FCC Rules.*

*Operation is subject to the following two conditions:*

- 1. This device may not cause harmful interference, and*
- 2. this device must accept any interference received, including interference that may cause undesired operation.*



# Document Revision History

- *Rev 3.2*
  - *Added separate Sony mode*
  - *Added warning to not exceed 5V on SWC input line*
- *Rev 3.1*
  - *Added troubleshooting content*
  - *Edited assembly instructions to include 5V power only spec*
- *Rev 3*
  - *Overhaul for RE\_SWC V2*
  - *Added disclaimers section*
  - *Added compliance section*
- *Rev 2 - FW V2.0.0*
  - *Added instructions for JVC headunit interface*
  - *Added instructions for Kenwood headunit interface*
- *Rev 1 - Initial release*